CLASS 10 MATHEMATICS :: SLIP TEST – FA-2

Maximum Marks: 20 • Time: 45 Minutes

Name: _____ **Roll No:** _____ **Sec:** _____ **Marks:** _____

SECTION I: Short Answer Questions — 3 × 2M = 6 Marks

1. Solve the following pair of linear equations by the elimination method: $2x + y = 7$ and $3x - 2y = 7$. **[2 M]**

2. Find the discriminant of $2x^2 - 5x + 3 = 0$ and hence determine the nature of its roots. **[2 M]**

3. Find the 18th term of the Arithmetic Progression: 4, 9, 14, 19, ... **[2 M]**

SECTION II: Long Answer Questions — 2 × 4M = 8 Marks

4. The sum of the first 7 terms of an Arithmetic Progression is 49 and that of 17 terms is 289. Find the first term (a), common difference (d), and calculate the sum of its first n terms (S_n). **[4 M]**

5. Calculate the Mode of the following grouped frequency distribution: **[4 M]**

Class Interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency	5	8	15	9	3

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Multiple Choice Section

SECTION III: Multiple Choice Questions **6 × 1M = 6 Marks**

6. If the lines given by $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are parallel, then the value of k is:

(A) −5/4 (B) 15/4 (C) 4/15 (D) 5/2

7. The roots of the quadratic equation $2x^2 - x - 6 = 0$ are:

(A) 2, −3/2 (B) −2, 3/2 (C) 2, 3/2 (D) −2, −3/2

8. Which term of the AP 21, 18, 15, ... is 0?

(A) 6th (B) 7th (C) 8th (D) 9th

9. The common difference of the AP where $a_{25} - a_{12} = -52$ is:

(A) 4 (B) −4 (C) −3 (D) 3

10. For a given distribution, Mode = 35 and Mean = 35. The Median of the distribution is:

(A) 30 (B) 35 (C) 40 (D) 70

11. In the modal formula $\text{Mode} = l + [(f_1 - f_0)/(2f_1 - f_0 - f_2)] \times h$, f_0 denotes:

(A) Frequency of modal class (B) Frequency of succeeding class (C) Frequency of preceding class (D) Cumulative frequency

CLASS 10 MATHEMATICS :: SOLUTIONS – FA-2

Section I & Section II Solutions

SECTION I: Short Answer Solutions — 6 Marks

Solution 1 (Linear System) — 2 M

$2x + y = 7$ ... (1) | $3x - 2y = 7$ ... (2)

Multiply (1) by 2: $4x + 2y = 14$

Add (2): $(4x + 2y) + (3x - 2y) = 14 + 7 \Rightarrow 7x = 21 \Rightarrow x = 3$

Substitute $x = 3$ into (1): $2(3) + y = 7 \Rightarrow 6 + y = 7 \Rightarrow y = 1$

$x = 3$ | $y = 1$

Solution 2 (Discriminant) — 2 M

$a = 2, b = -5, c = 3$

$D = b^2 - 4ac = (-5)^2 - 4(2)(3)$

$= 25 - 24 = 1$

$D = 1 > 0 \Rightarrow$ Two distinct real roots exist.

Solution 3 (nth Term of AP) — 2 M

$a = 4, d = 9 - 4 = 5, n = 18$

$a_{18} = a + (n - 1)d = 4 + (18 - 1)(5)$

$= 4 + 17(5) = 4 + 85 = 89$

18th term = 89

SECTION II: Long Answer Solutions — 8 Marks

Solution 4 (Sum of AP) — 4 M

$S_7 = (7/2)[2a + 6d] = 49 \Rightarrow 7(a + 3d) = 49 \Rightarrow a + 3d = 7$ ... (1)

$S_{17} = (17/2)[2a + 16d] = 289 \Rightarrow 17(a + 8d) = 289 \Rightarrow a + 8d = 17$ ... (2)

Subtract (1) from (2): $(a + 8d) - (a + 3d) = 17 - 7 \Rightarrow 5d = 10 \Rightarrow d = 2$

From (1): $a + 3(2) = 7 \Rightarrow a + 6 = 7 \Rightarrow a = 1$

$S_n = (n/2)[2(1) + (n - 1)(2)] = (n/2)[2 + 2n - 2] = (n/2)(2n) = n^2$

$a = 1$ | $d = 2$ | $S_n = n^2$

Solution 5 (Grouped Mode) — 4 M

Highest frequency = 15 $\Rightarrow$ Modal class = 20 – 30

$l = 20, f_1 = 15, f_0 = 8, f_2 = 9, h = 10$

$\text{Mode} = l + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$

$= 20 + [(15 - 8) / (2(15) - 8 - 9)] \times 10$

$= 20 + [7 / (30 - 17)] \times 10 = 20 + [7 / 13] \times 10 = 20 + 70/13 = 20 + 5.38$

Mode = 25.38

CLASS 10 MATHEMATICS :: SOLUTIONS – FA-2

Section III Multiple Choice Solutions

SECTION III: Multiple Choice Solutions — **6 Marks**

Solution 6 — **1 M**

Parallel condition: $a_1/a_2 = b_1/b_2$

$3/2 = (2k)/5 \Rightarrow 4k = 15 \Rightarrow k = 15/4$

Answer: (B) 15/4

Solution 7 — **1 M**

$2x^2 - 4x + 3x - 6 = 0$

$2x(x - 2) + 3(x - 2) = 0 \Rightarrow (x - 2)(2x + 3) = 0$

$x = 2, -3/2$

Answer: (A) 2, −3/2

Solution 8 — **1 M**

$a = 21, d = 18 - 21 = -3$

$21 + (n - 1)(-3) = 0 \Rightarrow -3(n - 1) = -21$

$n - 1 = 7 \Rightarrow n = 8$

Answer: (C) 8th

Solution 9 — **1 M**

$a_{25} - a_{12} = (a + 24d) - (a + 11d)$

$13d = -52 \Rightarrow d = -4$

Answer: (B) −4

Solution 10 — **1 M**

Empirical relation: Mode = 3 Median − 2 Mean

$35 = 3$ Median $- 2(35) \Rightarrow 3$ Median $= 105$

Median = 35

Answer: (B) 35

Solution 11 — **1 M**

In the modal formula, f_0 is the frequency of the preceding class.

Answer: (C) Frequency of preceding class